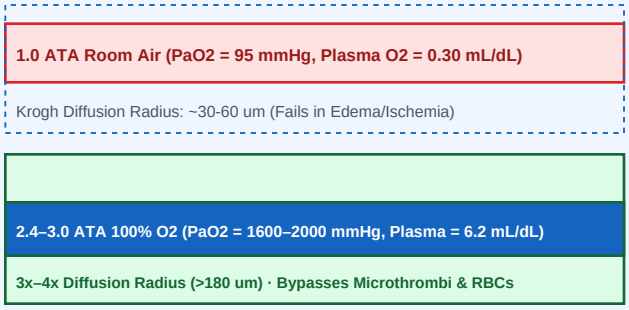
THE DEFINITIVE BOOK OF
HYPERBARIC CHAMBER THERAPY

GAS PHYSICS · MOLECULAR REGENERATION · ASME PVHO-1 ENGINEERING · CLINICAL PROTOCOLS

FIG 1.1: QUANTITATIVE BIOPHYSICS OF HYPERBARIC OXYGENATION (1.0 ATA NORMOBARIC VS. 2.4–3.0 ATA HBOT)



GOVERNING EQUATIONS OF HYPERBARIC MEDICINE

1. Alveolar Gas Equation:

```
PAO2 = FiO2 * (760 * P_ATA - 47) - PaCO2 / R
```

2. Henry's Law (Plasma O2):

```
C_plasma = 0.0031 mL O2/(dL*mmHg) * PaO2
```

3. Total Arterial O2 Content:

```
CaO2 = (1.34 * Hgb * SaO2) + (0.0031 * PaO2)
```

4. Boyle's Law (Bubble/Volume):

```
P1 * V1 = P2 * V2  (At 2.8 ATA, V2 = 35.7% V1)
```

5. Pulmonary Toxic Dose (UPTD):

```
UPTD = t * ((PO2 - 0.5) / 0.5)^0.83  [Max 615/day]
```

EXECUTIVE RESEARCH SYNTHESIS: This compendium provides the complete scientific, clinical, engineering, and regulatory architecture required to write the definitive textbook on Hyperbaric Chamber Therapy. Every section distinguishes **VERIFIED** clinical/regulatory facts (16 UHMS 15th Edition indications, 15 FDA 510(k) cleared uses, 15 CMS Medicare NCD 20.29 covered indications, ASME PVHO-1 / NFPA 99 Chapter 14 engineering codes) from **VERIFIED RCT OFF-LABEL** protocols (post-stroke neuroplasticity, mTBI/post-concussion, Long COVID, PTSD, fibromyalgia, ulcerative colitis) and **PROPOSED / REFUTED** marketing claims.

PART I: HISTORY & GAS BIOPHYSICS (Ch. 1–4)

From Henshaw's 1662 Domicilium, Fontaine's 1879 mobile OR, and Cunningham's 1928 Steel Ball Hospital to Boerema's 1960 'Life Without Blood' (3.0 ATA). Full derivation of Boyle's, Dalton's, Henry's, Graham's, and Fick's laws, Krogh diffusion cylinders, and UPTD dosing kinetics.

PART II: MOLECULAR & CELLULAR MECHANISMS (Ch. 5–8)

The Hyperoxic-Hypoxic Paradox (HHP): how intermittent hyperoxia-normoxia fluctuations stabilize HIF-1a, HIF-2a, VEGF, and SIRT1; Thom's 8-fold CD34+ eNOS stem cell mobilization; neutrophil beta-2 integrin inhibition; mitochondrial biogenesis, PBMC telomere elongation (>20%), and senolysis.

PART III: CHAMBER ENGINEERING & CODES (Ch. 9–12)

ASME PVHO-1 pressure vessel engineering and NFPA 99 Chapter 14 hyperbaric facility codes. Class A multiplace steel chambers (BIBS masks, water deluge) vs. Class B monoplace acrylic chambers vs. 1.3 ATA soft-shell mHBOT bags. Complete UHMS / FDA / CMS NCD 20.29 regulatory matrix.

PART IV: UHMS/FDA APPROVED PROTOCOLS (Ch. 13–17)

Step-by-step protocols for all 16 UHMS 15th Edition indications: US Navy Table 6 (DCS/AGE), Weaver protocol for CO poisoning, Clostridial gas gangrene (3.0 ATA), NSTI, Marx 30/10 Osteoradionecrosis protocol, Wagner III+ Diabetic Foot Ulcers (TcPO2 > 200 mmHg), CRAO, ISSHL, and Femoral Head AVN.

PART V: EMERGING & OFF-LABEL FRONTIERS (Ch. 18–21)

Double-blind RCT evidence in Post-COVID Condition (PMID: 35821512, 38360929), Chronic Post-Stroke & TBI Neuroplasticity (PMID: 23335971, 24260334), Veteran PTSD (PMID: 35192645), Fibromyalgia, Ulcerative Colitis (NCT05987852), Oncology radiosensitization, and resolution of the 1.3 ATA Sham Paradox.

PART VI: PHARMACOLOGY, SAFETY & OPERATIONS (Ch. 22–24)

Untreated pneumothorax (sole absolute contraindication), chemotherapeutic/enzymatic interactions (Doxorubicin, Bleomycin, Cisplatin, Disulfiram, Mafenide Acetate, Insulin hypoglycemia), Teed Middle Ear Barotrauma grading, Paul Bert CNS seizure management, and Pre-Dive Checklists.

SECTION 1: COMPLETE 24-CHAPTER MANUSCRIPT BLUEPRINT (CHAPTERS 1–12)

CH 01: From Henshaw's Domicilium to 'Life Without Blood': 360 Years of Hyperbaric Med
[Boerema 1960 (J Cardiovasc Surg), PMID: 34577787]

Chronicles the evolution of hyperbaric therapy from Nathaniel Henshaw's 1662 organ-bellows domicilium and 19th-century European pneumatic institutes (Junod 1834, Pravaz 1837, Fontaine's 1879 wheeled hyperbaric surgical amphitheater) through Paul Bert's 1878 discovery of CNS oxygen toxicity and Lorrain Smith's 1899 pulmonary toxicity, Orval Cunningham's 1928 64-foot 'Steel Ball Hospital' in Cleveland, Behnke & Shaw's 1937 US Navy decompression tables, and Ite Boerema's landmark 1956-1960 'Life Without Blood' cardiac surgery experiments in Amsterdam that established modern evidence-based hyperbaric medicine.

CH 02: Fundamental Gas Laws & Hyperbaric Biophysics: Boyle, Dalton, Henry, Graham & F
[PMID: 30538529, PMID: 33642335]

Derives the five physical laws governing hyperbaric medicine: Boyle's Law (P1V1 = P2V2) explaining both bubble volume crushing in DCS/AGE and barotrauma in closed cavities; Dalton's Law of Partial Pressures; Henry's Law (C = kH * P) governing linear plasma oxygen dissolution (0.0031 mL O2/dL/mmHg); Graham's Law of gas effusion; and Fick's First Law coupled with the Krogh-Erlanger Cylinder Model demonstrating how 2.0-3.0 ATA expands the radial oxygen diffusion distance from capillary endothelium from ~30 um to >100-200 um across edematous, micro-thrombosed tissue.

CH 03: Alveolar-Arterial Oxygen Transport & Hemoglobin-Independent Oxygenation
[Boerema 1960, PMID: 34577787]

Provides the complete quantitative physiology of oxygen transport from the Alveolar Gas Equation (PAO2 = FiO2*(PB - PH2O) - PaCO2/R) to arterial blood content (CaO2 = 1.34*Hgb*SaO2 + 0.0031*PaO2). Proves why normobaric 100% O2 cannot exceed ~1.86 mL/dL dissolved O2 once hemoglobin is 100% saturated, whereas 3.0 ATA 100% O2 achieves PAO2 = 2,193 mmHg and ~6.2 mL/dL dissolved plasma O2—exceeding resting human arteriovenous oxygen extraction (~5.5 mL/dL) without a single red blood cell.

CH 04: Oxygen as a Precision Drug: Dosing Variables, Air Breaks & UPTD Kinetics
[PMID: 32630465, PMID: 31409407]

Establishes the pharmacological dosing matrix of HBOT (Pressure in ATA, Duration in minutes, FiO2, Intermittent Air Breaks, and Cumulative Session Count). Details the Unit Pulmonary Toxic Dose equation UPTD = t * ((PO2 - 0.5)/0.5)^0.83 (daily safe limit <= 615 UPTD; extreme emergency limit <= 1,425 UPTD) and proves how 5-minute normoxic air breaks every 20-30 minutes both prevent CNS oxygen seizures and drive regenerative signaling.

CH 05: The Hyperoxic-Hypoxic Paradox (HHP): HIF-1alpha, SIRT1 & Epigenetic Signaling
[PMID: 32630465, PMID: 36613522]

Dissects the paradigm shift discovered by Efrati, Hadanny, and Balestra: why intermittent hyperoxia followed by rapid return to normoxia tricks cellular oxygen sensors into perceiving relative hypoxia. During hyperoxia, Upregulated antioxidant scavengers (SOD, glutathione peroxidase, catalase) outpace ROS when PO2 drops during air breaks or decompression, stabilizing HIF-1a/HIF-2a, upregulating VEGF, EPO, and SIRT1 neuroprotection without ischemic damage.

CH 06: Endogenous Stem Cell Mobilization, eNOS & Microvascular Neovascularization
[PMID: 32066777, PMID: 29097988]

Examines Stephen R. Thom's discovery that a single 2.0 ATA HBOT exposure doubles circulating CD34+/CD45-dim stem/progenitor cells and 20 treatments induce an 8-fold (800%) surge via bone marrow eNOS nitric oxide synthesis, matrix metalloproteinase-9 (MMP-9) activation, and soluble Kit ligand cleavage. Connects circulating progenitor homing with local VEGF/bFGF release and prolyl-4-hydroxylase collagen maturation.

CH 07: Immunomodulation, Neutrophil Beta-2 Integrin Blockade & Antimicrobial Action
[PMID: 16604253, PMID: 29761012]

Explains the dual immunological miracle of HBOT: (1) restoring neutrophil NADPH oxidase superoxide killing in hypoxic wounds (which ceases when tissue PO2 < 30 mmHg and peaks above 100-300 mmHg), and (2) selectively inhibiting neutrophil beta-2 integrin (CD11b/CD18) adhesion to endothelial ICAM-1 via S-nitrosylation, halting ischemia-reperfusion injury while suppressing NF-kB, TNF-alpha, IL-1beta, and IL-6.

CH 08: Mitochondrial Bioenergetics, Telomere Elongation & Cellular Senolysis
[PMID: 33206062, PMID: 34944468]

Analyzes mitochondrial biogenesis (PGC-1alpha, Complex I-IV respiration in athletes, PMID: 35133516) and the landmark 2020 Shamir Medical Center trial (Hachmo et al., PMID: 33206062) reporting >20% PBMC telomere elongation and up to 37% clearance of senescent CD28-null T cells after 60 HBOT sessions at 2.0 ATA. Includes rigorous scientific critique distinguishing true telomere elongation from bone-marrow mobilization of younger leukocyte subsets.

CH 09: Class A Multiplace vs. Class B Monoplace Hard-Shell Engineering (ASME PVHO-1 &
[NFPA 99 Ch. 14, ASME PVHO-1]

Comprehensive engineering guide to ASME PVHO-1 (Safety Standard for Pressure Vessels for Human Occupancy) and NFPA 99 Chapter 14. Compares Class A multiplace steel chambers (pressurized with compressed air to 2.0-6.0 ATA, patients breathing 100% O2 via BIBS masks/hoods with overboard exhaust keeping room O2 < 23.5%, inside medical tender access) against Class B monoplace cast-acrylic chambers (pressurized directly with 100% O2 up to 3.0 ATA, external control console, specialized hyperbaric IV pass-through penetrators and ventilators).

CH 10: Mild Hyperbaric Chambers (mHBOT, 1.3 ATA Soft-Shell) vs. Medical Grade Systems
[FDA Class II 510(k), NFPA 99 Ch. 14]

Dissects the clinical, engineering, and legal divide between portable soft-shell TPU/nylon zippered chambers (1.3 ATA / 4.4 psi gauge, ambient air or 24% O2 concentrator) and hard-shell medical chambers (1.5-3.0 ATA, 100% medical O2). Documents why FDA 510(k) clears soft chambers solely for Acute Mountain Sickness (AMS), calculates the 7x-10x difference in arterial PaO2 (135-160 mmHg vs. 1,300-2,000 mmHg), and warns against lethal fire hazards from connecting aftermarket oxygen concentrators to ungrounded fabric bags.

CH 11: Fire Physics in Oxygen-Enriched Atmospheres, Grounding & Deluge Systems
[NFPA 99 Ch. 14]

Examines fire ignition energy in 100% O2 at 2-3 ATA (where flame propagation speed increases 5x-10x and auto-ignition temperatures plummet). Details mandatory 100% cotton scrubs, zero-tolerance contraband lists (lithium batteries, smartphones, chemical hand-warmers, petroleum/vaseline, alcohol, hairspray), <10 Ohm static grounding verification, and NFPA 99 Class A high-velocity water deluge systems.

CH 12: The Regulatory & Reimbursement Matrix: UHMS 15th Ed., FDA 510(k) & CMS NCD 20.
[CMS NCD 20.29, UHMS 15th Ed.]

Presents the complete side-by-side crosswalk of the 16 UHMS 15th Edition approved indications, 15 FDA cleared indications, and 15 CMS Medicare NCD 20.29 covered conditions (plus the 15 explicitly non-covered CMS conditions). Details the Wagner Grade III diabetic foot ulcer documentation rules and Transcutaneous Oximetry (TcPO2) provocative testing (<40 mmHg room air, >200 mmHg in-chamber).

SECTION 2: COMPLETE 24-CHAPTER MANUSCRIPT BLUEPRINT (CHAPTERS 13–24)

CH 13: Diving & Critical Care Emergencies: Decompression Sickness, Gas Embolism & CO
[PMID: 25906396, NCT04490317]

Step-by-step clinical execution of US Navy Treatment Tables 5, 6, and 6A for Decompression Sickness and Arterial Gas Embolism (including delayed presentation >48h, PMID: 25906396), and the Weaver NEJM protocol (3.0 ATA initial dive + two 2.0 ATA dives within 24h) for acute Carbon Monoxide and Cyanide poisoning to prevent Delayed Neurological Sequelae (DNS).

CH 14: Life-Threatening Infections: Gas Gangrene, Necrotizing Fasciitis, Brain Absces
[PMID: 34577787, PMID: 23162231]

Protocols for surgical-hyperbaric integration in Clostridium perfringens myonecrosis (3.0 ATA TID Day 1), Fournier's gangrene and polymicrobial necrotizing fasciitis (2.0-2.5 ATA BID), anaerobic intracranial abscesses, and Cierny-Mader IIIB/IVB chronic refractory osteomyelitis.

CH 15: Delayed Radiation Injuries: Marx 30/10 Protocol, Osteoradionecrosis & Soft Tis
[NCT07140198, NCT05189496]

Complete guide to Robert Marx's 30/10 prophylactic dental extraction protocol and 20/10 mandibular reconstruction staging at 2.4 ATA, plus protocols for radiation cystitis, radiation proctitis, laryngeal radionecrosis, and irradiated breast reconstruction (NCT07140198, NCT05189496).

CH 16: Diabetic Foot Ulcers, Compromised Flaps/Grafts, Crush Injuries, Burns & Blood
[PMID: 20427683, PMID: 23423696]

Reconciles the landmark Londahl 2010 RCT (PMID: 20427683, 52% vs 29% complete ulcer healing at 1 year) and Sharma 2021 meta-analysis (PMID: 33500533) with the Margolis 2013 observational study (PMID: 23423696), demonstrating why TcPO2 selection and Wagner III+ protocol adherence determine limb salvage. Covers failing flap salvage (2.0-2.5 ATA BID), acute crush injuries, thermal burns, and severe blood loss anemia.

CH 17: Neurosensory & Orthopedic Emergencies: CRAO, Sudden Hearing Loss & Femoral Hea
[PMID: 28096655, NCT06128720]

Details emergency stroke-code HBOT for Central Retinal Artery Occlusion within the <12-24h window before irreversible retinal ganglion death (Hadanny et al. 2017, PMID: 28096655; Stanford optic neuropathy trial NCT06128720), Idiopathic Sudden Sensorineural Hearing Loss within 14 days, and the newest UHMS 15th Edition indication: Steinberg Stage I-II Avascular Necrosis of the Femoral Head.

CH 18: Neurorehabilitation: TBI, Post-Concussion Syndrome, Post-Stroke & The Sham Con
[PMID: 24260334, PMID: 23335971]

Synthesizes the prospective RCTs and neuroimaging studies in chronic post-stroke neuroplasticity (Efrati 2013 PMID: 23335971; Hadanny 2020 PMID: 31985478), mild TBI and post-concussion syndrome in adults and children (Boussi-Gross 2013 PMID: 24260334; Tal 2017 PMID: 29097988; Hadanny 2022 PMID: 36151105), anoxic brain injury (PMID: 26409406), and acute stroke thrombectomy trials (NCT07049692, NCT07626450, BioHOBIT NCT04855396). Resolves the DoD military mTBI sham controversy by proving that 1.2-1.3 ATA compressed air elevates PaO2 by >35% and is an active low-dose pressurization arm, not a true placebo.

CH 19: Long COVID, Treatment-Resistant PTSD & Central Sensitization (Fibromyalgia)
[PMID: 35821512, PMID: 38360929]

Examines double-blind sham-controlled RCTs demonstrating microstructural brain repair (DTI/fMRI) and clinical symptom resolution in Post-COVID Condition (Zilberman-Itskovich 2022 PMID: 35821512; 1-year follow-up Hadanny 2024 PMID: 38360929; Toronto RCT NCT06452095), combat-veteran PTSD (Doenyas-Barak 2022 PMID: 35192645; IDF RCT NCT07473362), and Fibromyalgia (Efrati 2015 PMID: 26010952; Hadanny 2018 PMID: 30618929; Ablin 2023 PMID: 36897850).

CH 20: Inflammatory Bowel Disease, Oncology Adjuncts, Athletic Performance & Healthy
[PMID: 23054400, PMID: 27485098]

Covers UHMS 15th Ed. emerging indication Inflammatory Bowel Disease (Ulcerative Colitis HBOT-UC RCT NCT05987852 & perianal Crohn's), Oncology safety and radiosensitization (Moen & Stuhr 2012 PMID: 23054400; Stepien 2016 PMID: 27485098; Brizel 1995 PMID: 7577456; Poff 2013 PMID: 23755243; NCT06811870), middle-aged athlete mitochondrial VO2max RCT (PMID: 35133516), and healthy older adult cognitive enhancement RCT (Hadanny 2020 PMID: 32589613).

CH 21: Scientific Boundaries: Where Evidence Refutes Efficacy or Demands Consumer Pro
[NCT00584480, FDA Consumer Warnings]

Provides an unflinching truth-first evaluation of conditions where RCTs show negative or inconclusive results (Multiple Sclerosis, Autism Spectrum Disorder NCT00584480, Cerebral Palsy where 1.3 ATA air equaled oxygen arms) and echoes FDA warnings against predatory commercial clinics marketing soft-shell bags as cures for cancer, HIV, or advanced neurodegenerative disease.

CH 22: Contraindications, Drug Interactions & Implanted Device Clearance
[PMID: 34577787, PMID: 28760643]

Complete safety pharmacology reference covering Untreated Pneumothorax (absolute contraindication), Doxorubicin, Bleomycin, Cisplatin, Disulfiram, Mafenide Acetate, Insulin/Hypoglycemia dynamics, and ASME/manufacturer pressure certification for pacemakers, ICDs, cochlear implants, and intrathecal pumps.

CH 23: Barotrauma (Teed Grading), Ocular Myopia/Cataracts & CNS/Pulmonary Oxygen Toxi
[PMID: 28760643, PMID: 31409407]

Clinical prevention and acute management of Middle Ear Barotrauma (Teed 0-5, NCT05697328), sinus/dental/pulmonary barotrauma, reversible hyperoxic myopia vs. nuclear cataracts (McMonnies 2018 PMID: 28760643), pulmonary function stability under standard 60-session protocols (Hadanny 2019 PMID: 31409407), and emergency management of Paul Bert grand mal oxygen seizures (VENTID prodrome; never decompress during tonic phase).

CH 24: Facility Operations, Credentialing, Pre-Dive Checklists & Next-Generation HBOT
[UHMS Accreditation Manual, NFPA 99 Ch. 14]

Operational blueprint for hyperbaric medicine centers: ABMS Undersea & Hyperbaric Medicine physician subspecialty boarding, CHT/CHRN staffing, UHMS accreditation, turn-key Pre-Dive Safety Checklists, In-Chamber Emergency Action Cards, and 2026-2035 technological horizons (closed-loop EEG/TcPO2 titration and multi-omics response prediction).

SECTION 3: 360 YEARS OF HYPERBARIC EVOLUTION & QUANTITATIVE ALVEOLAR-ARTERIAL TABLE

1662	**Nathaniel Henshaw (UK)**	Built the 'Domicilium' using organ bellows to alter ambient air pressure for pulmonary/digestive therapy.
1834–1879	**Junod, Pravaz & Fontaine (France)**	Pneumatic institutes across Europe; Fontaine (1879) built a wheeled hyperbaric operating room (27 surgeries).
1878 & 1899	**Paul Bert & J. Lorrain Smith**	Bert discovered CNS O2 toxicity seizures ('Paul Bert Effect'); Smith discovered pulmonary O2 toxicity.
1928–1942	**Orval Cunningham (Cleveland, OH)**	Constructed the 64-ft, 5-story 'Steel Ball Hospital'; AMA censure for unproven claims; scrapped for WWII steel.
1937	**Behnke & Shaw (US Navy)**	Introduced hyperbaric oxygen for decompression sickness nitrogen washout; foundation of US Navy Table 6.
1956–1960	**Ite Boerema (Amsterdam)**	'Life Without Blood' (3.0 ATA): kept hemoglobin-depleted swine alive solely via 6.2 mL/dL dissolved plasma O2.
1961–1985	**Brummelkamp, UHMS & R. Marx**	Gas gangrene alpha-toxin arrest (1961); UHMS founded (1967); Marx 30/10 Osteoradionecrosis protocol (1985).
2002–2026	**Weaver, Thom, Efrati & Hadanny**	Weaver CO RCT (2002); Thom 8x CD34+ stem cells (2006); Hyperoxic-Hypoxic Paradox, Long COVID & AVN (2020-2026).

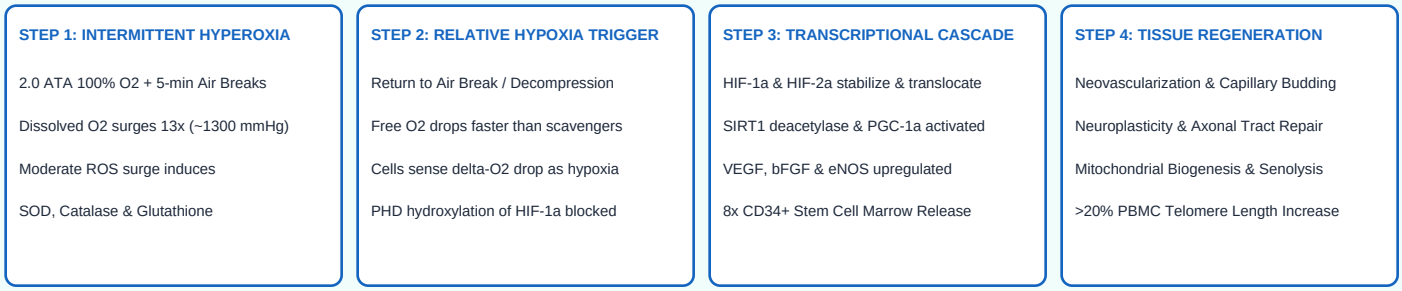
TABLE 4.1: QUANTITATIVE GAS PHYSICS & ARTERIAL OXYGEN DISSOLUTION (1.0 ATA TO 3.0 ATA)

ATA / Depth	FiO2 & Modality	PAO2 (mmHg)	PaO2 (mmHg)	Plasma O2	Total CaO2	Clinical Tier
1.0 ATA (0 fsw)	21% · Room Air (Normobaric)	102 mmHg	**95 mmHg**	**0.30 mL/dL**	**20.1 mL/dL**	Baseline Physiology
1.0 ATA (0 fsw)	100% · 100% O2 Mask (Normobaric	673 mmHg	**600 mmHg**	**1.86 mL/dL**	**21.9 mL/dL**	Emergency First Aid O2
1.3 ATA (10 fsw)	21% · Compressed Air (mHBOT So	148 mmHg	**135 mmHg**	**0.42 mL/dL**	**20.4 mL/dL**	Mild Hyperbaric Air (Bio
1.3 ATA (10 fsw)	24% · mHBOT Soft Bag + O2 Conc	176 mmHg	**160 mmHg**	**0.50 mL/dL**	**20.6 mL/dL**	Mild HBOT (FDA Cleared f
1.5 ATA (16.5 fsw)	100% · 100% Medical O2 (Hard Ch	1053 mmHg	**950 mmHg**	**2.95 mL/dL**	**23.0 mL/dL**	Neuro-Regenerative Proto
2.0 ATA (33 fsw)	100% · 100% Medical O2 (Hard Ch	1433 mmHg	**1300 mmHg**	**4.03 mL/dL**	**24.1 mL/dL**	Standard Wound, Neuro-Ag
2.4 ATA (46 fsw)	100% · 100% Medical O2 (Hard Ch	1737 mmHg	**1600 mmHg**	**4.96 mL/dL**	**25.1 mL/dL**	Marx Osteoradionecrosis
2.8 ATA (60 fsw)	100% · 100% Medical O2 (US Navy	2041 mmHg	**1900 mmHg**	**5.89 mL/dL**	**26.0 mL/dL**	Decompression Sickness &
3.0 ATA (66 fsw)	100% · 100% Medical O2 (Maximum	2193 mmHg	**2000 mmHg**	**6.20 mL/dL**	**26.3 mL/dL**	Boerema 'Life Without Bl

CRITICAL BIOPHYSICAL INSIGHT — THE 1.3 ATA 'SHAM' PARADOX: Notice in Row 3 above that compressing a chamber to just **1.3 ATA on room air (21% O2)** elevates arterial PaO2 from 95 mmHg to **135 mmHg (+42%)** and increases dissolved plasma O2 to 0.42 mL/dL. In Department of Defense (DoD) mTBI trials and cerebral palsy studies that used 1.2–1.3 ATA compressed air as a 'sham placebo control,' both the 1.3 ATA air arm and the HBOT arm experienced significant clinical improvement over non-pressurized standard care—proving 1.3 ATA air is a biologically active low-dose pressurization intervention, whereas true inert sham requires 1.03 ATA or 1.3 ATA with nitrogen-diluted 16% O2 to hold PaO2 constant at 95 mmHg.

SECTION 4: THE HYPEROXIC-HYPOXIC PARADOX (HHP) & REGENERATIVE CASCADES

FIG 5.1: THE HYPEROXIC-HYPOXIC PARADOX SIGNALING CASCADE (HADANNY & EFRATI, PMID: 32630465)



1. HIF-1alpha, VEGF & Angiogenesis Across the '3-H' Radiation/Ischemic Bed (PMID: 32630465, 32066777)

In chronic wounds and delayed radiation injury, tissue is trapped in a hypocellular-hypovascular-hypoxic ('3-H') state where baseline PO2 (<15 mmHg) is too low to support fibroblast collagen synthesis (which requires PO2 >= 30-40 mmHg for prolyl-4-hydroxylase activity). HBOT creates a steep 1,500+ mmHg oxygen gradient across the wound margin and triggers intermittent HHP cycles during 5-minute air breaks. This stabilizes HIF-1a and upregulates Vascular Endothelial Growth Factor (VEGF), basic Fibroblast Growth Factor (bFGF), and nitric oxide (Yamamoto et al. 2020, PMID: 32066777), restoring microvascular density to ~80% of normal non-irradiated tissue.

2. Endogenous Bone Marrow CD34+ Stem Cell Mobilization via eNOS (Thom 2006; Ortega 2021 PMID: 34577787)

Hyperbaric oxygen stimulates medullary endothelial Nitric Oxide Synthase (eNOS) within the bone marrow parenchyma. The resulting burst of nitric oxide (NO) S-nitrosylates and activates Matrix Metalloproteinase-9 (MMP-9), which cleaves membrane-bound Kit ligand into soluble Kit ligand (sKitL / stem cell factor). This releases hematopoietic and endothelial progenitor cells (CD34+/CD45-dim/VEGFR-2+) into systemic circulation—doubling after a single 2.0 ATA treatment and increasing 8-fold (800%) after 20 treatments. These mobilized progenitors home directly to SDF-1/CXCR4 gradients in ischemic myocardium, diabetic ulcers, and injured brain tissue.

3. Neutrophil Beta-2 Integrin Inhibition vs. Enhanced Leukocyte Bacterial Killing (PMID: 16604253, 29761012)

HBOT resolves a critical immunological paradox: in infected hypoxic wounds, neutrophil NADPH oxidase cannot generate bactericidal superoxide/hypochlorous acid when PO2 < 30 mmHg; HBOT restores PO2 > 200 mmHg, amplifying bacterial phagocytosis and killing by >400%. Simultaneously, in acute ischemia-reperfusion (crush injury, flap salvage, CO poisoning), HBOT inhibits neutrophil beta-2 integrin (CD11b/CD18) clustering via membrane guanylate cyclase/cGMP signaling, preventing neutrophils from adhering to endothelial ICAM-1 and halting microvascular reperfusion destruction without causing systemic immunosuppression.

4. Mitochondrial Respiration, Telomere Elongation & Immunosenescence Clearance (PMID: 33206062, 34944468, 35133516)

Schottlender et al. (2021, PMID: 34944468) and Hadanny et al. (2022 RCT in middle-aged athletes, PMID: 35133516) demonstrated that 40–60 sessions of intermittent HBOT (2.0 ATA) upregulate PGC-1alpha and SIRT1, increasing maximal mitochondrial phosphorylation capacity and VO2max. In the landmark prospective aging trial by Hachmo, Hadanny & Efrati (2020, Aging, PMID: 33206062), 60 daily HBOT sessions (2.0 ATA, 90 min with three 5-min air breaks) elongated telomeres by >20% across B cells (+25–37%), helper T cells (+29%), and natural killer cells (+22%), while reducing senescent CD28-null T-helper cells by 37.3%. Author's critical note: the definitive book must present both the direct telomere-protective mechanisms and the hematological discussion regarding whether PBMC telomere shifts partially reflect efflux of younger bone-marrow-derived leukocyte cohorts.

SECTION 5: UHMS 15TH EDITION, FDA 510(k) & CMS MEDICARE NCD 20.29 PROTOCOLS (PART A: 1–8)

#01. AIR OR GAS EMBOLISM (AGE)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.8 - 3.0 ATA (US Navy Table 6) · **Duration:** 285 min (Table 6 with air breaks) · **Course:** 1 - 3 (up to 5-10 until neurological plateau)

Mechanism & Clinical Rationale: Boyle's Law mechanical bubble volume compression (64% volume reduction at 2.8 ATA), nitrogen washout via steepened diffusion gradient, and reduction of leukocyte-mediated endothelial injury/cerebral edema.

#02. CARBON MONOXIDE POISONING (INCLUDING COMPLICATED BY CYANIDE)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.5 - 3.0 ATA (Weaver Protocol: 3.0 ATA then 2.0 ATA) · **Duration:** 90 - 120 min · **Course:** 1 - 3 sessions within 24 hours

Mechanism & Clinical Rationale: Accelerates carboxyhemoglobin (COHb) dissociation (t1/2 drops from 300 min on room air to ~20 min at 3.0 ATA), restores cytochrome c oxidase (Complex IV), blocks brain lipid peroxidation, and halves Delayed Neurological Sequelae (DNS).

#03. CLOSTRIDIAL MYONECROSIS (GAS GANGRENE)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 3.0 ATA · **Duration:** 90 min · **Course:** TID on Day 1, then BID for 4-5 days (total 5-10)

Mechanism & Clinical Rationale: Achieves tissue PO2 > 250 mmHg, immediately halting Clostridium perfringens alpha-toxin (lecithinase) production, demarcating viable tissue to reduce amputation level alongside surgical debridement and antibiotics.

#04. CRUSH INJURIES, COMPARTMENT SYNDROME & ACUTE TRAUMATIC ISCHEMIAS

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.0 - 2.4 ATA · **Duration:** 90 min · **Course:** BID for 2-3 days, then daily (total 6-12)

Mechanism & Clinical Rationale: Hyperoxic vasoconstriction reduces interstitial edema by 20-30% while plasma hyperoxygenation maintains microcirculatory O2 delivery and blocks beta-2 integrin neutrophil reperfusion injury.

#05. DECOMPRESSION SICKNESS (DCS TYPE I & II)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.8 ATA (US Navy Table 5 or Table 6) · **Duration:** 135 min (Table 5) to 285+ min (Table 6) · **Course:** 1 - 5 sessions (Hadanny et al. 2015, PMID: 25906396 shows benefit even with delayed recompression >48h)

Mechanism & Clinical Rationale: Crushes intravascular and extravascular inert gas bubbles via Boyle's Law, creates zero-inspired-N2 washout gradient, and reverses spinal/cerebral venous infarction.

#06. DIABETIC FOOT ULCERS & SELECTED PROBLEM WOUNDS

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED (Strictly Wagner Grade >= III after 30 days failed standard care)

Dosing Protocol: 2.0 - 2.4 ATA · **Duration:** 90 - 120 min · **Course:** 30 - 40 daily sessions (re-evaluated every 30 days)

Mechanism & Clinical Rationale: Elevates wound TcPO2 > 200 mmHg, stimulates HIF-1a/VEGF angiogenesis, mobilizes CD34+ stem cells, and powers fibroblast collagen cross-linking and leukocyte bacterial killing (Londahl 2010 PMID: 20427683; Sharma 2021 PMID: 33500533).

#07. EXCEPTIONAL BLOOD LOSS ANEMIA

UHMS: APPROVED | FDA: CLEARED | CMS: LOCAL MAC DISCRETION

Dosing Protocol: 2.0 - 3.0 ATA · **Duration:** 90 - 120 min (intermittent as required by ischemic symptoms) · **Course:** TID/BID until bone marrow erythropoiesis restores Hgb

Mechanism & Clinical Rationale: Boerema 'Life Without Blood' principle: dissolves up to 6.0-6.8 mL O2/dL directly into plasma, meeting resting metabolic O2 consumption in patients who cannot receive transfusions (e.g. Jehovah's Witnesses or severe hemolytic crossmatch incompatibility).

#08. INTRACRANIAL ABSCESS (CIA)

UHMS: APPROVED | FDA: CLEARED | CMS: LOCAL MAC DISCRETION

Dosing Protocol: 2.0 - 2.5 ATA · **Duration:** 60 - 90 min · **Course:** BID initially, then daily (10 - 20 sessions)

Mechanism & Clinical Rationale: Inhibits anaerobic flora dominating cerebral/subdural/epidural abscesses, reduces perifocal cerebral edema and ICP via hyperoxic vasoconstriction, and enhances blood-brain barrier antibiotic penetration.

SECTION 6: UHMS 15TH EDITION, FDA 510(k) & CMS MEDICARE NCD 20.29 PROTOCOLS (PART B: 9–16)

#09. NECROTIZING SOFT TISSUE INFECTIONS (NSTI / FOURNIER'S GANGRENE)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.0 - 2.5 ATA · **Duration:** 90 min · **Course:** BID for 3-5 days, then daily (10 - 20 sessions)

Mechanism & Clinical Rationale: Restores neutrophil NADPH oxidase respiratory burst in hypoxic fascial planes, potentiates aminoglycosides/vancomycin/fluoroquinolones, and demarcates necrotic tissue.

#10. CHRONIC REFRACTORY OSTEOMYELITIS (CIERNY-MADER IIIB / IVB)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.0 - 2.4 ATA · **Duration:** 90 - 120 min · **Course:** 30 - 40 daily sessions

Mechanism & Clinical Rationale: Raises infected bone oxygen tension from <15 mmHg to >100 mmHg, activating osteoclast resorption of necrotic sequestra, polymorphonuclear bacterial killing, and osteogenesis.

#11. DELAYED RADIATION INJURIES (SOFT TISSUE & OSTEORADIONECROSIS)

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED

Dosing Protocol: 2.0 - 2.4 ATA · **Duration:** 90 - 120 min · **Course:** Marx 30/10 Protocol (30 pre-op + 10 post-op) or 30 - 60 sessions

Mechanism & Clinical Rationale: Reverses obliterative endarteritis and the '3-H' (hypocellular-hypovascular-hypoxic) radiation bed by creating steep O2 gradients across irradiated margins that drive capillary budding (restoring vascular density to ~80% of normal).

#12. COMPROMISED SKIN GRAFTS AND FLAPS

UHMS: APPROVED | FDA: CLEARED | CMS: COVERED (Preparation/preservation of compromised grafts)

Dosing Protocol: 2.0 - 2.5 ATA · **Duration:** 90 - 120 min · **Course:** BID for 3-5 days, then daily (total 10 - 20)

Mechanism & Clinical Rationale: Sustains ischemic graft/flap viability via plasmatic imbibition, prevents ischemia-reperfusion neutrophil plug formation, and accelerates inosculation and neovascularization.

#13. ACUTE THERMAL BURNS (>20% TBSA SECOND/THIRD DEGREE)

UHMS: APPROVED | FDA: CLEARED | CMS: NON-COVERED (Explicitly excluded in NCD 20.29 B.4)

Dosing Protocol: 2.0 - 2.4 ATA · **Duration:** 90 min · **Course:** BID for first 3-5 days (within 24h of burn), then daily (10 - 30)

Mechanism & Clinical Rationale: Preserves ATP in the zone of stasis (preventing conversion of partial-thickness to full-thickness burns), reduces fluid resuscitation volume via hyperoxic vasoconstriction, and accelerates re-epithelialization.

#14. IDIOPATHIC SUDDEN SENSORINEURAL HEARING LOSS (ISSHL)

UHMS: APPROVED (Added 2011) | FDA: CLEARED | CMS: NON-COVERED (Commercial payer dependent)

Dosing Protocol: 2.0 - 2.5 ATA · **Duration:** 90 min · **Course:** 10 - 20 daily sessions (initiate within 14 days of onset; max 3 months)

Mechanism & Clinical Rationale: Cochlear organ of Corti lacks direct vasculature and relies solely on diffusion from perilymph/endolymph; HBOT elevates perilymphatic PO2 by >450%, rescuing ischemic hair cells alongside systemic/intratympanic steroids.

#15. CENTRAL RETINAL ARTERY OCCLUSION (CRAO)

UHMS: APPROVED | FDA: CLEARED | CMS: NON-COVERED (Emergency hospital protocol)

Dosing Protocol: 2.0 - 2.8 ATA · **Duration:** 90 - 120 min · **Course:** BID until retinal artery recanalization (must initiate within <12-24h while cherry-red spot/viable inner retina remains; Hadanny 2017 PMID: 28096655)

Mechanism & Clinical Rationale: Oxygenates the ischemic inner retinal layers via diffusion from the intact choroidal circulation (which normally supplies only the outer 1/3 of the retina, but supplies ~100% of retinal thickness at >2.0 ATA O2).

#16. AVASCULAR NECROSIS (AVN / OSTEONECROSIS) OF THE FEMORAL HEAD

UHMS: APPROVED (Newest UHMS 15th Edition Indication) | FDA: OFF-LABEL / PENDING | CMS: NON-COVERED

Dosing Protocol: 2.0 - 2.5 ATA · **Duration:** 90 min · **Course:** 40 - 60 daily sessions (Steinberg Stage I - II pre-collapse)

Mechanism & Clinical Rationale: Reduces intraosseous hypertension and bone marrow edema, downregulates RANKL/osteoclast bone resorption while upregulating osteoprotegerin (OPG) and Wnt/beta-catenin osteoblast bone repair.

SECTION 7: HYPERBARIC CHAMBER ENGINEERING, CODES & FIRE SAFETY ARCHITECTURE

CLASS A: MULTIPLACE STEEL
2.0 – 6.0 ATA · ASME PVHO-1

- Occupancy: 2-20+ patients + tender
- Vessel: Steel hull + PVHO-1 viewports
- Chamber Gas: Air (O2 strictly <23.5%)
- Breathing: 100% O2 via BIBS/hoods
- Fire Code: NFPA 99 water deluge
- Edge: Hands-on critical care access
- Room: 2-hr fire walls + BIBS exhaust

CLASS B: MONOPLACE ACRYLIC
1.5 – 3.0 ATA · ASME PVHO-1

- Occupancy: Single supine patient
- Vessel: Cast PMMA acrylic cylinder
- Chamber Gas: 100% Medical Grade O2
- Breathing: Ambient O2 + air-break mask
- Fire Code: Zero-spark (<10 Ohm ground)
- Edge: High throughput, 360° visibility
- ICU: Hyperbaric IVs & ventilator ports

MILD SOFT-SHELL (mHBOT)
1.3 ATA (4.4 psi) · NON-PVHO-1

- Occupancy: Zippered TPU/nylon bag
- Vessel: Flexible polymer bladder
- Chamber Gas: Room air (21% O2)
- Breathing: Air or 24-30% concentrator
- PaO2: 135-160 mmHg (vs 1300-2000)
- FDA 510(k): Acute Mountain Sickness
- Hazard: Fire risk if 100% O2 piped in

ASME PVHO-1 Acrylic Viewport Integrity & Lifecycle Governance

All human-occupied clinical hyperbaric chambers in North America must carry an ASME Boiler and Pressure Vessel Code Section VIII stamp coupled with an ASME PVHO-1 (Pressure Vessels for Human Occupancy) data plate. Because cast polymethyl methacrylate (PMMA) acrylic undergoes viscoelastic creep, cyclic fatigue, and UV/chemical crazing over thousands of compression cycles, ASME PVHO-1 mandates strict viewport thickness-to-diameter ratios, prohibition of alcohol/solvent cleaners (which induce instant stress-cracking), annual visual craze inspections, and formal engineering recertification at the 10-year / 20,000-cycle design life milestone.

NFPA 99 Chapter 14: The Hyperbaric Fire Triangle & Contraband Protocol

In a Class B chamber filled with 100% oxygen at 2.0–3.0 ATA, the minimum ignition energy drops by two orders of magnitude and flame propagation velocity increases 5- to 10-fold, rendering any internal fire virtually unsurvivable. NFPA 99 Chapter 14 enforces the Three Pillars of Hyperbaric Fire Prevention: (1) Elimination of Ignition Sources: patient wears 100% cotton UHMS-approved garments, wears conductive grounding strap verified <10 Ohms to facility earth ground, and undergoes a 2-person pre-dive pocket/body frisk prohibiting lithium-ion batteries, phones, smartwatches, hearing aids, chemical hand-warmers (iron-oxidation exothermic packs caused fatal chamber fires), and synthetic/silk/wool fabrics; (2) Elimination of Fuels: strict ban on petroleum jelly (Vaseline), oil-based cosmetics, lip balm, alcohol-based deodorants/hairspray, and titanium dioxide sunscreens; (3) Class A Deluge & Overboard Dump: Class A multiplace chambers continuously monitor ambient O2 (alarming at >=23.5% O2), vent exhaled BIBS oxygen outside the building, and maintain an automatic + manual water deluge system capable of flooding the chamber within 3 seconds.

Transcutaneous Oximetry (TcPO2) Provocative Testing Protocol

To predict whether an ischemic lower-extremity wound (e.g., Wagner Grade III+ diabetic foot ulcer) will respond to HBOT, periwound Transcutaneous Oxygen Monitoring (TcPO2) is the gold-standard physiological assay: (1) Baseline Room Air TcPO2 < 40 mmHg confirms severe tissue hypoxia; (2) Normobaric 100% O2 challenge evaluates macrovascular inflow; (3) In-Chamber TcPO2 at 2.0–2.4 ATA rising to >= 200 mmHg proves sufficient collateral microvascular patency to achieve a >80% probability of limb salvage and wound closure.

SECTION 8: CONTRAINDICATIONS, PHARMACOLOGY & ADVERSE EVENT MANAGEMENT

UNTREATED PNEUMOTHORAX — [ABSOLUTE CONTRAINDICATION]
CRITICAL / FATAL RISK

Mechanism: Boyle's Law (P1V1 = P2V2): During decompression from 2.0-3.0 ATA to 1.0 ATA, trapped pleural air expands 2x to 3x in volume, rapidly converting a simple pneumothorax into a fatal tension pneumothorax with cardiovascular collapse.
Clinical Protocol: STRICTLY PROHIBITED until a tube thoracostomy (chest tube) with a Heimlich valve or water-seal drainage is placed and lung re-expansion is radiographically confirmed.

DOXORUBICIN (ADRIAMYCIN) — [Pharmacological Interaction (Chemotherapeutic)]
HIGH RISK (Cardiotoxicity)

Mechanism: Anthracycline quinone redox cycling generates superoxide and hydroxyl radicals in cardiomyocytes (which lack high catalase levels). Concurrent HBOT massively amplifies myocardial oxidative stress and fatal congestive heart failure.
Clinical Protocol: Discontinue doxorubicin at least 72 hours (3-7 days minimum washout) prior to initiating HBOT.

BLEOMYCIN — [Pharmacological Interaction (Chemotherapeutic)]
HIGH / MODERATE RISK (Pulmonary Fibrosis)

Mechanism: Bleomycin forms complexes with Fe2+ and O2 that cleave DNA and induce alveolar pneumocyte inflammation and irreversible interstitial pulmonary fibrosis (hydrolase deficiency in lung tissue).
Clinical Protocol: Historically considered a lifetime contraindication; modern UHMS guidelines permit HBOT with caution if >6 months post-exposure, cumulative dose <300,000 IU (400 mg), normal DLCO/PFTs, and high-res chest CT clearance.

CISPLATIN (CIS-PLATINUM) — [Pharmacological Interaction (Chemotherapeutic)]
MODERATE RISK (Impaired Wound Healing)

Mechanism: Inhibits fibroblast proliferation, DNA synthesis, and collagen cross-linking, blunting the angiogenic and wound-healing efficacy of HBOT.
Clinical Protocol: Delay elective wound-healing HBOT until cisplatin course is completed; emergency life/limb-saving HBOT (e.g. CO poisoning, gas gangrene) is NOT withheld.

DISULFIRAM (ANTABUSE) — [Pharmacological Interaction (Enzyme Inhibitor)]
HIGH RISK (CNS & Pulmonary O2 Toxicity)

Mechanism: Directly inhibits Cu/Zn Superoxide Dismutase (SOD), stripping the central nervous system and lungs of their primary antioxidant defense against hyperbaric superoxide radicals, precipitating Paul Bert grand mal seizures.
Clinical Protocol: Discontinue disulfiram prior to multi-session HBOT; never administer concurrent HBOT at >=2.0 ATA without washout.

MAFENIDE ACETATE (SULFAMYLON) — [Pharmacological Interaction (Topical Burn Antimicrobial)]
MODERATE RISK (Vasoconstriction & Acidosis)

Mechanism: Carbonic anhydrase inhibitor that induces local CO2 retention, metabolic acidosis, and peripheral vasoconstriction, counteracting HBOT microvascular tissue oxygenation.
Clinical Protocol: Thoroughly wipe and cleanse all mafenide acetate cream from burn wounds prior to chamber entry (additionally prevents petroleum/excipient fire hazard).

INSULIN & SULFONYLUREAS — [Metabolic / Endocrine Interaction]
HIGH FREQUENCY (Chamber Hypoglycemia)

Mechanism: HBOT acutely increases peripheral insulin sensitivity and GLUT4 glucose uptake, causing blood glucose to drop by 25 to 50+ mg/dL during a 90-minute dive; hypoglycemia both mimics and lowers the threshold for CNS oxygen toxicity seizures.
Clinical Protocol: Mandatory pre-dive fingerstick blood glucose check for all diabetic patients (target >= 120-150 mg/dL before compression); keep oral glucose gel accessible.

CLINICAL EMERGENCY CARDS: MIDDLE EAR BAROTRAUMA (TEED SCALE) & CNS OXYGEN SEIZURE (VENTID)

1. Modified Teed Classification of Middle Ear Barotrauma (MEB, Incidence 2–17%): Grade 0: Symptoms without otoscopic signs · **Grade 1:** Diffuse redness/retraction of Shrapnell's membrane & manubrium · **Grade 2:** Erythema of entire tympanic membrane (TM) · **Grade 3:** Hemorrhage within TM substance · **Grade 4:** Hemotympanum / free blood in middle ear · **Grade 5:** Tympanic membrane perforation. *Prevention:* Slow compression rate (1–2 psi/min), active Valsalva/Frenzel equalization, or myringotomy/tympanostomy tubes in intubated patients.

2. Paul Bert CNS Oxygen Toxicity Seizure Protocol (Incidence ~1–4 per 10,000 dives at >=2.0 ATA): VENTID Prodrome: Vision (tunnel vision), Ears (tinnitus), Nausea, Twitching (facial/diaphragmatic myoclonus), Irritability/anxiety, Dizziness. **IMMEDIATE ACTION:** (a) Immediately switch patient breathing gas from 100% O2 to compressed air (remove BIBS mask or apply air mask); seizures typically self-terminate within 60–90 seconds once brain PO2 drops (note: transient 'off-effect' seizure continuation for 15–30s after switching to air is physiological); (b) **GOLDEN RULE: NEVER DECOMPRESS THE CHAMBER DURING THE TONIC PHASE OF A SEIZURE!** During the tonic phase, the glottis is spasmed shut; decompressing the chamber against a closed glottis causes catastrophic pulmonary barotrauma, pneumothorax, and fatal cerebral Arterial Gas Embolism (AGE). Wait until rhythmic respiration resumes before slow decompression.

SECTION 9: EMERGING FRONTIERS & ACTIVE CLINICAL TRIALS (373 TOTAL REGISTERED TRIALS)

EVIDENCE STRATIFICATION IN OFF-LABEL & TRANSLATIONAL HBOT: A definitive textbook must rigorously separate **high-quality prospective double-blind RCTs** from unsubstantiated commercial claims:

• **Post-COVID Condition (Long COVID) [VERIFIED DOUBLE-BLIND RCT]:** Zilberman-Itskovich et al. (2022, *Sci Rep*, PMID: 35821512, N=73) and Hadanny et al. (2024 1-year follow-up, PMID: 38360929) demonstrated significant restoration of global cognition, executive function, attention, energy, sleep, pain, and DTI/fMRI frontoparietal perfusion following 40 sessions at 2.0 ATA (90 min with 5-min air breaks).

• **Chronic Post-Stroke & Mild TBI / Post-Concussion Syndrome [VERIFIED PROSPECTIVE RCTs]:** Efrati et al. (2013, PMID: 23335971) and Boussi-Gross et al. (2013, PMID: 24260334) proved via SPECT/DTI imaging that metabolically 'stunned' penumbral neurons remain viable years after stroke or concussion and can be reactivated via 40–60 sessions at 1.5–2.0 ATA.

• **Treatment-Resistant PTSD & Fibromyalgia [VERIFIED RCTs]:** Doenyas-Barak et al. (2022, PMID: 35192645) in combat veterans and Efrati et al. (2015, PMID: 26010952) / Ablin et al. (2023, PMID: 36897850) in post-TBI/trauma fibromyalgia demonstrated neuroplastic reversal of limbic/prefrontal hyperexcitability.

• **Oncology Safety & Radiosensitization [VERIFIED REVIEWS & PHASE I/II]:** Moen & Stuhr (2012, PMID: 23054400) and Stepien et al. (2016, PMID: 27485098) confirmed across decades of clinical data that HBOT does **not** promote malignancy growth or metastasis, and functions as a potent radiosensitizer in hypoxic glioblastomas.

TABLE 10.1: REPRESENTATIVE CLINICALTRIALS.GOV HBOT STUDIES (RETRIEVED VIA CLINICAL_TRIALS_API.PY)

NCT ID	Status / Phase	N	Lead Sponsor	Official Trial Title
NCT00584480	COMPLETED (PHASE3)	10	University of California	Pilot Study of the Effect of Hyperbaric Oxygen T
NCT06493851	RECRUITING (NA)	40	Assaf-Harofeh Medical Ce	The Effect of Hyperbaric Oxygen Therapy for Acce
NCT07140198	RECRUITING (NA)	16	University Hospital, Gen	Hyperbaric Oxygen Therapy to Improve Autologous
NCT06811870	RECRUITING (PHASE1,)	161	Guangdong Provincial Peo	A Phase II Single-Arm Clinical Study to Explore
NCT06452095	RECRUITING (NA)	120	University Health Networ	Recovering From COVID-19 Lingering Symptoms Adap
NCT07678073	RECRUITING (NA)	68	University of Gaziantep	Comparison of the Effects of General Anesthesia
NCT07430150	RECRUITING (NA)	100	Assaf-Harofeh Medical Ce	Long Lasting Effect of Hyperbaric Oxygen Therapy
NCT03594344	RECRUITING (NA)	120	Oslo University Hospital	Additional Hyperbaric Oxygen After Lower Extremi
NCT07049692	RECRUITING (NA)	424	Beijing Chao Yang Hospit	Hyperbaric Oxygen After Stroke Thrombectomy: A M
NCT04490317	RECRUITING (NA)	1500	Wonju Severance Christia	CARbon monoxidE intoxiCatiOn in Korea: Prospecti
NCT05189496	RECRUITING (NA)	380	Guangdong Provincial Peo	Hyperbaric Oxygen in the Prevention of Radiation
NCT05987852	RECRUITING (NA)	126	Northwestern University	Hyperbaric Oxygen Therapy for Ulcerative Colitis
NCT05842239	RECRUITING (NA)	15	Assaf-Harofeh Medical Ce	Hyperbaric Oxygen Therapy for Men Suffering From
NCT07626450	RECRUITING (NA)	420	Beijing Tiantan Hospital	HBO After Thrombectomy for AIS With LVO
NCT04791488	RECRUITING (NA)	265	Direction Centrale du Se	Impact of Hyperoxia and Involvement of the Immun
NCT07662629	RECRUITING (NA)	40	Cairo University	Addtional Effect of Hyperbaric Oxygen to Program
NCT05591300	RECRUITING (NA)	108	Azienda Ospedaliero Univ	Microparticles Blood Level in Acute Carbon Monox
NCT05697328	RECRUITING (NA)	90	John Muir Health	Barotrauma in Hyperbaric Oxygen Therapy
NCT02280733	RECRUITING (NA)	300000	U.S. Wound Registry	A Real-World Registry of Chronic Wounds and Ulce
NCT06128720	RECRUITING (NA)	60	Stanford University	Hyperbaric Oxygen Therapy for Optic Neuropathies
NCT07198412	RECRUITING (NA)	180	Benha University	Effect of Sedation After Ultrasound-Guided Spina

SECTION 10: VERIFIED OPEN-ACCESS LITERATURE BIBLIOGRAPHY (EUROPE PMC — PART 1)

[01] PMID: 20427683 | PMC2858204 | Year: 2010 | Citations: 261 | DOI: 10.2337/dc09-1754 https://europepmc.org/article/MED/20427683
Löndahl M, Katzman P, Nilsson A, Hamma — Hyperbaric oxygen therapy facilitates healing of chronic foot ulcers in patien (Diabetes care)

[02] PMID: 23054400 | PMC3510426 | Year: 2012 | Citations: 215 | DOI: 10.1007/s11523-012-0233-x https://europepmc.org/article/MED/23054400
Moen I, Stuhr LE. — Hyperbaric oxygen therapy and cancer--a review. (Targeted oncology)

[03] PMID: 34577787 | PMC8465921 | Year: 2021 | Citations: 195 | DOI: 10.3390/medicina57090864 https://europepmc.org/article/MED/34577787
Ortega MA, Fraile-Martinez O, García-M — A General Overview on the Hyperbaric Oxygen Therapy: Applications, Mechanisms (Medicina (Kaunas, Lithuan)

[04] PMID: 32630465 | PMC7355982 | Year: 2020 | Citations: 167 | DOI: 10.3390/biom10060958 https://europepmc.org/article/MED/32630465
Hadanny A, Efrati S. — The Hyperoxic-Hypoxic Paradox. (Biomolecules)

[05] PMID: 24260334 | PMC3829860 | Year: 2013 | Citations: 143 | DOI: 10.1371/journal.pone.0079995 https://europepmc.org/article/MED/24260334
Boussi-Gross R, Golan H, Fishlev G, Be — Hyperbaric oxygen therapy can improve post concussion syndrome years after mil (PloS one)

[06] PMID: 23755243 | PMC3673985 | Year: 2013 | Citations: 137 | DOI: 10.1371/journal.pone.0065522 https://europepmc.org/article/MED/23755243
Poff AM, Ari C, Seyfried TN, D'Agostin — The ketogenic diet and hyperbaric oxygen therapy prolong survival in mice with (PloS one)

[07] PMID: 23335971 | PMC3546039 | Year: 2013 | Citations: 130 | DOI: 10.1371/journal.pone.0053716 https://europepmc.org/article/MED/23335971
Efrati S, Fishlev G, Bechor Y, Volkov — Hyperbaric oxygen induces late neuroplasticity in post stroke patients--random (PloS one)

[08] PMID: 34944468 | PMC8699286 | Year: 2021 | Citations: 111 | DOI: 10.3390/biom11121827 https://europepmc.org/article/MED/34944468
Schottlender N, Gottfried I, Ashery U. — Hyperbaric Oxygen Treatment: Effects on Mitochondrial Function and Oxidative S (Biomolecules)

[09] PMID: 28760643 | PMC5777925 | Year: 2018 | Citations: 105 | DOI: 10.1016/j.optom.2017.06.002 https://europepmc.org/article/MED/28760643
McMonnies C. — Reactive oxygen species, oxidative stress, glaucoma and hyperbaric oxygen ther (Journal of optometry)

[10] PMID: 23423696 | PMC3687310 | Year: 2013 | Citations: 98 | DOI: 10.2337/dc12-2160 https://europepmc.org/article/MED/23423696
Margolis DJ, Gupta J, Hoffstad O, Papd — Lack of effectiveness of hyperbaric oxygen therapy for the treatment of diabet (Diabetes care)

[11] PMID: 23299244 | PMC3587810 | Year: 2013 | Citations: 95 | DOI: 10.1038/jcbfm.2012.179 https://europepmc.org/article/MED/23299244
Yan W, Fang Z, Yang Q, Dong H, Lu Y, L — SirT1 mediates hyperbaric oxygen preconditioning-induced ischemic tolerance in (Journal of cerebral blood)

[12] PMID: 35821512 | PMC9276805 | Year: 2022 | Citations: 94 | DOI: 10.1038/s41598-022-15565-0 https://europepmc.org/article/MED/35821512
Zilberman-Itskovich S, Catalogna M, Sa — Hyperbaric oxygen therapy improves neurocognitive functions and symptoms of po (Scientific reports)

[13] PMID: 32066777 | PMC7026099 | Year: 2020 | Citations: 94 | DOI: 10.1038/s41598-020-59615-x https://europepmc.org/article/MED/32066777
Yamamoto N, Oyaizu T, Enomoto M, Horie — VEGF and bFGF induction by nitric oxide is associated with hyperbaric oxygen-i (Scientific reports)

[14] PMID: 23162231 | PMC3495382 | Year: 2012 | Citations: 93 | DOI: 10.4103/0970-0358.101309 https://europepmc.org/article/MED/23162231
Bhutani S, Vishwanath G. — Hyperbaric oxygen and wound healing. (Indian journal of plastic)

[15] PMID: 16604253 | PMC5917171 | Year: 2006 | Citations: 90 | DOI: 10.1100/tsw.2006.78 https://europepmc.org/article/MED/16604253
Al-Waili NS, Butler GJ. — Effects of hyperbaric oxygen on inflammatory response to wound and trauma: pos (TheScientificWorldJournal)

[16] PMID: 27485098 | PMC4971045 | Year: 2016 | Citations: 89 | DOI: 10.1007/s12032-016-0814-0 https://europepmc.org/article/MED/27485098
St■pie■ K, Ostrowski RP, Matyja E. — Hyperbaric oxygen as an adjunctive therapy in treatment of malignancies, inclu (Medical oncology (Northwo)

[17] PMID: 34680155 | PMC8533945 | Year: 2021 | Citations: 87 | DOI: 10.3390/biom11101520 https://europepmc.org/article/MED/34680155
Gottfried I, Schottlender N, Ashery U. — Hyperbaric Oxygen Treatment-From Mechanisms to Cognitive Improvement. (Biomolecules)

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Choudhury R. — Hypoxia and hyperbaric oxygen therapy: a review. (International journal of)

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Brizel DM, Lin S, Johnson JL, Brooks J — The mechanisms by which hyperbaric oxygen and carbogen improve tumour oxygenat (British journal of cancer)

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Tal S, Hadanny A, Sasson E, Suzin G, E — Hyperbaric Oxygen Therapy Can Induce Angiogenesis and Regeneration of Nerve Fi (Frontiers in human neuros)

[21] PMID: 26010952 | PMC4444341 | Year: 2015 | Citations: 72 | DOI: 10.1371/journal.pone.0127012 https://europepmc.org/article/MED/26010952
Efrati S, Golan H, Bechor Y, Faran Y, — Hyperbaric oxygen therapy can diminish fibromyalgia syndrome--prospective clin (PloS one)

[22] PMID: 33500533 | PMC7838311 | Year: 2021 | Citations: 71 | DOI: 10.1038/s41598-021-81886-1 https://europepmc.org/article/MED/33500533
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Hentia C, Rizzato A, Camporesi E, Yang — An overview of protective strategies against ischemia/reperfusion injury: The (Brain and behavior)

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TURN-KEY PRE-DIVE HYPERBARIC SAFETY & CLINICAL CLEARANCE CHECKLIST (NFPA 99 / UHMS STANDARD)

[1] Pulmonary & Air-Trapping Clearance: Recent Chest X-Ray/CT verified negative for **untreated pneumothorax** or severe bullous emphysema. **[2] Pharmacological Audit:** Verified zero active **Doxorubicin** (<72h), **Disulfiram**, **Cisplatin**, or topical **Mafenide Acetate**; historical **Bleomycin** screened with PFT/DLCO. **[3] Otoscopic & Equalization Check:** Baseline Teed Grade 0; patient demonstrated Valsalva/Frenzel maneuver or tympanostomy tubes patent. **[4] Endocrine & Vital Signs:** Temperature < 38.5°C (fever lowers CNS O2 seizure threshold); Diabetic fingerstick glucose >= 120–150 mg/dL. **[5] NFPA 99 Two-Person Contraband Frisk:** 100% cotton scrubs; zero lithium batteries, phones, watches, hearing aids, chemical hand-warmers, lighters, petroleum/Vaseline, lip balm, alcohol, or hairspray; wrist grounding strap verified < 10 Ohms.